csd5748

ΗΥ-111: Απειροστικός Λογισμός ΙΙ: Άσκηση 4η

1η Άσκηση:

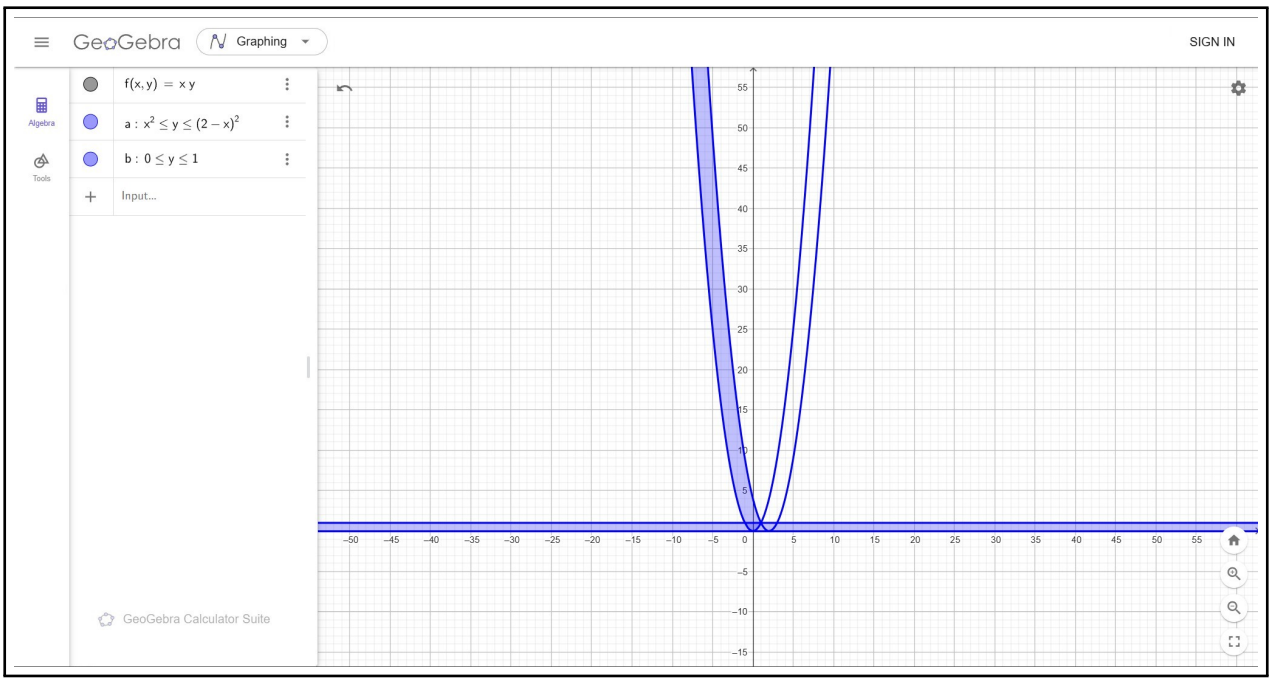
$\alpha)$ Δεδομένα: $\displaystyle\int_0^1 \int_{\sqrt{y}}^{2-\sqrt{y}} xy\ dxdy \ = \ \int_0^1 \left(\int_{\sqrt{y}}^{2-\sqrt{y}} (xy)\ dx \right) dy$

Θέτω $I = \displaystyle\int_{\sqrt{y}}^{2-\sqrt{y}} xy\ dx$ και έχουμε $\displaystyle\int_0^1 (I)\ dy$

$$I = y \int_{\sqrt{y}}^{2-\sqrt{y}} (x)\, dx = y \left[\frac{x^2}{2} \right]_{\sqrt{y}}^{2-\sqrt{y}} = y \left(\frac{(2-\sqrt{y})^2}{2} - \frac{y}{2} \right) = y \cdot \frac{4-4\sqrt{y}}{2} = y(2-2\sqrt{y}) = 2y - 2y\sqrt{y}$$

$$\text{Άρα} \quad \int_0^1 (I)\ dy = \int_0^1 (2y - 2y\sqrt{y})\ dy = 2\int_0^1 (y)\ dy - 2\int_0^1 (y\sqrt{y})\ dy = 2\left[\frac{y^2}{2} \right]_0^1 - 2\left[\frac{y^{5/2}}{5/2} \right]_0^1 = 2 \cdot \frac{1}{2} - 2 \cdot \frac{2}{5} = \frac{1}{5}$$

Τελικά: $\boxed{\displaystyle\int_0^1 \int_{\sqrt{y}}^{2-\sqrt{y}} xy\ dxdy = \frac{1}{5}}$



∗ Χωρίο Ολοκλήρωσης - Σχεδιασμένο στο Geogebra ...

$\beta)$ Δεδομένα: $\displaystyle\int_0^1\int_0^{x^3} e^{y/x}\,dydx \;=\; \int_0^1\left(\int_0^{x^3}(e^{y/x})\,dy\right)dx$

Θέτω $\displaystyle I \;=\; \int_0^{x^3} e^{y/x}\,dy$ και έχουμε $\displaystyle\int_0^1 (I)\,dx$

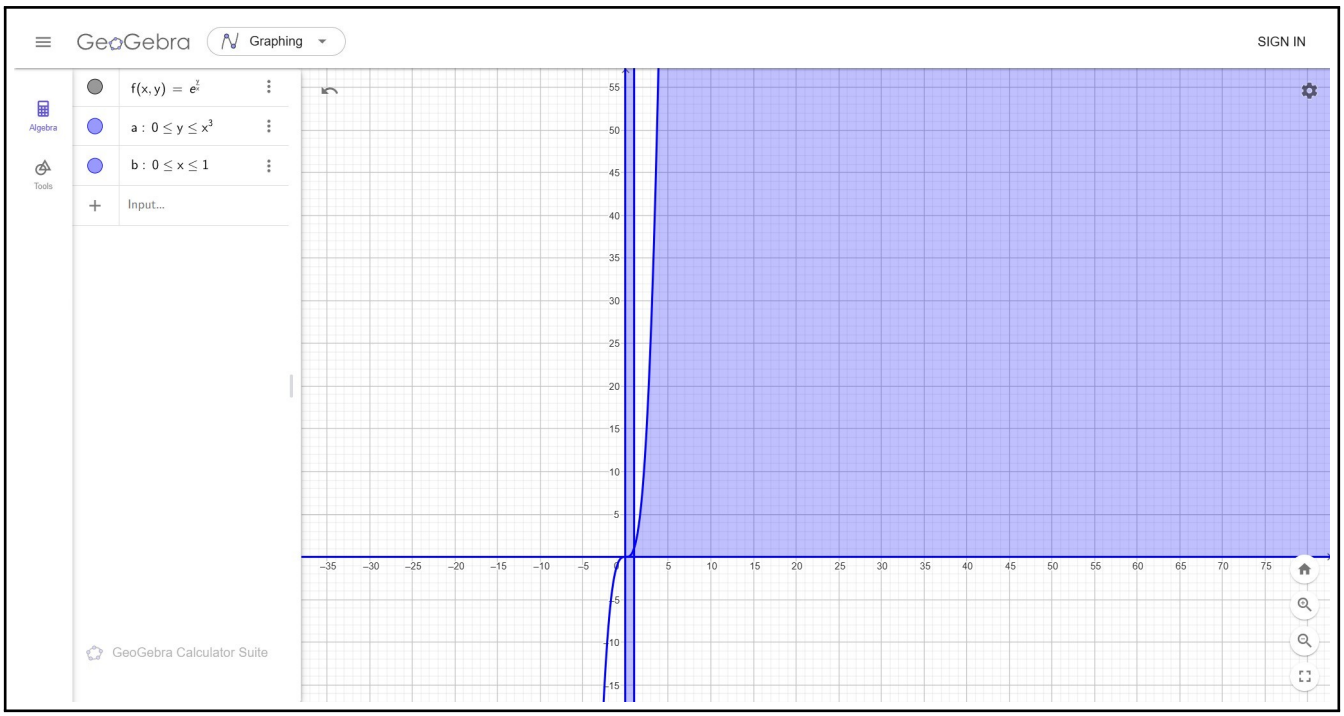
Θέτω $u=\dfrac{y}{x}\Rightarrow u'=\dfrac{du}{dy}\Rightarrow dy=x\,du$ και $y=0\Rightarrow u=0,\; y=x^3\Rightarrow u=x^2$ και έχουμε:

$$I \;=\; \int_0^{x^2}(e^u\cdot x)\,du = x\int_0^{x^2}(e^u)\,du \;=\; x\cdot[e^u]_0^{x^2} = x(e^{x^2}-1)$$

Άρα $\displaystyle\int_0^1 (I)\,dx \;=\; \int_0^1 x(e^{x^2}-1)\,dx = \int_0^1 (xe^{x^2})\,dx - \int_0^1 (x)\,dx$ $\quad\begin{array}{c} w=x^2\Rightarrow w'=\frac{dw}{dx}\Rightarrow dx=\frac{1}{2x}dw \\ \hline x=0\Rightarrow w=0\;\;\&\;\; x=1\Rightarrow w=1\end{array}$

$$\Longrightarrow \int_0^1 (e^w\cdot\frac{x}{2x})\,dw - \int_0^1 (x)\,dx = \frac{1}{2}\,[e^w]_0^1 - \left[\frac{x^2}{2}\right]_0^1 = \frac{1}{2}(e-1)-\frac{1}{2} = \frac{e-2}{2}$$

Τελικά: $\boxed{\displaystyle\int_0^1\int_0^{x^3} e^{y/x}\,dydx = \frac{1}{2}(e-2)}$



∗ Χωρίο Ολοκλήρωσης - Σχεδιασμένο στο Geogebra ...

$2^{η}$ Άσκηση:

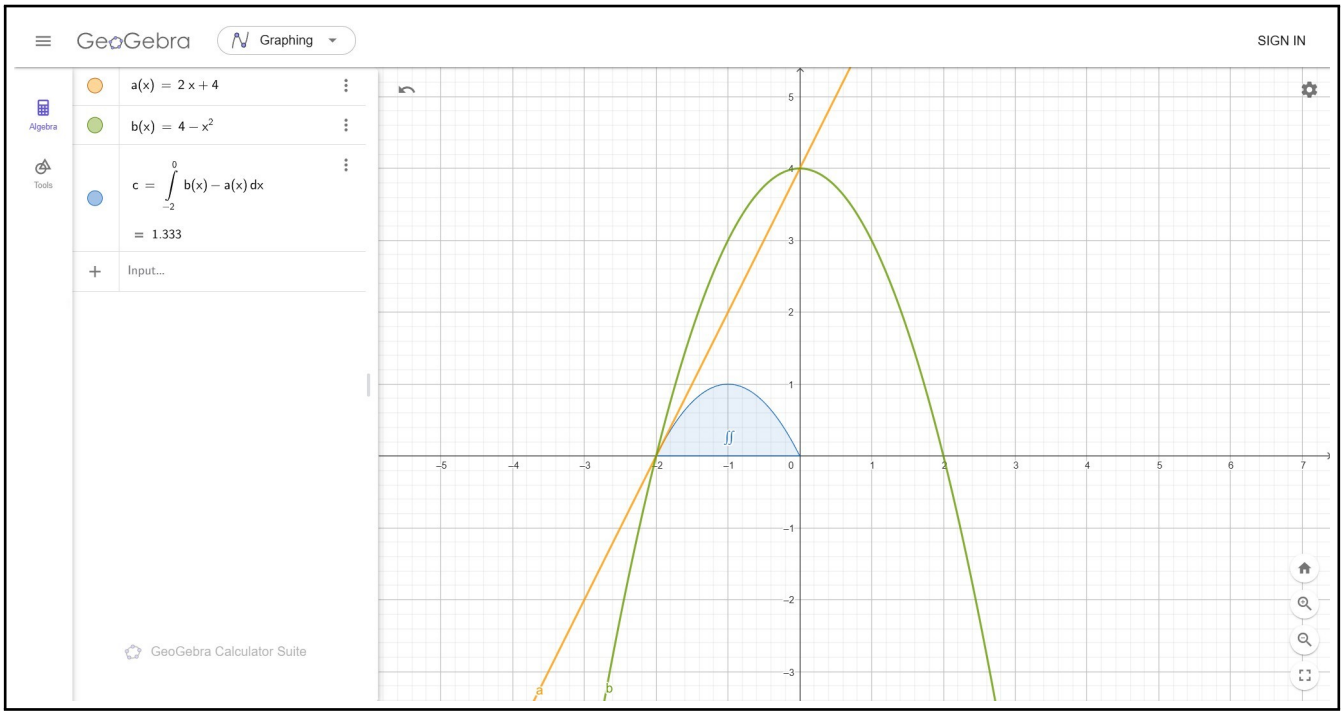
$\alpha)$ Δεδομένα: παρ.: $y = 4 - x^2$ και ε: $y = 2x + 4$

Σημεία Τομής: $4 - x^2 = 2x + 4 \Rightarrow x(x+2) = 0 \Rightarrow x = 0$ ή $x = -2 \Rightarrow y = 4$ & $y = 0 \Rightarrow \begin{cases} \Sigma_1 = (0,4) \\ \Sigma_2 = (-2,0) \end{cases}$

Για ολοκλήρωση ως προς $x \in [-2, 0] \longrightarrow y = 4 - x^2 > y = 2x + 4$.

$$\Longrightarrow \int_{-2}^{0} ((4 - x^2) - (2x + 4))\, dx = \int_{-2}^{0} (-x^2 - 2x)\, dx = \left[-\frac{x^3}{3} - x^2 \right]_{-2}^{0} = \cdots = \frac{4}{3}$$

Τελικά: $\boxed{\text{Εμβαδόν} = \frac{4}{3} \text{ μονάδες}}$



∗ Χωρίο Ολοκλήρωσης - Σχεδιασμένο στο Geogebra ...

$\beta)$ Δεδομένα: $\int_0^2 \int_0^{4-x^2} 2x\ dy\ dx = \int_0^2 (\int_0^{4-x^2} (2x)\ dy\ )\ dx$

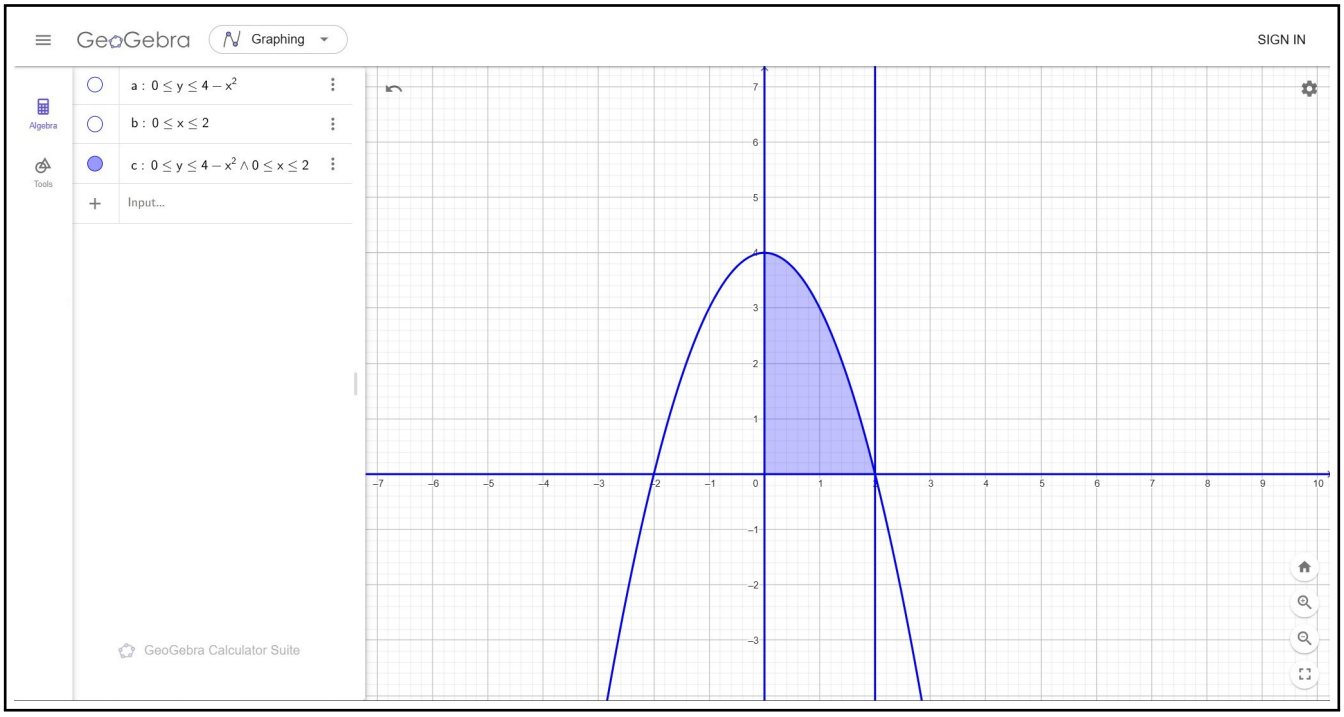
Έχουμε: $\int_0^{4-x^2} dy \longrightarrow \left\{ \begin{array}{c} y = 4-x^2 \\ y = 0 \end{array} \right. \Longrightarrow \left\{ \begin{array}{c} x = \pm\sqrt{4-y} \\ y = 0 \end{array} \right. \underset{[0,2]}{\overset{x\ \in}{\Longrightarrow}} \left\{ \begin{array}{c} x = \sqrt{4-y} \\ y = 0 \end{array} \right. \longrightarrow \int_0^{\sqrt{4-y}} dx$

Επιπλέον: $\int_0^2 dx \longrightarrow \left\{ \begin{array}{l} x = 0 \Rightarrow y = 4 - 0^2 = 4 \\ x = 2 \Rightarrow y = 4 - 2^2 = 0 \end{array} \right. \longrightarrow \int_0^4 dy$

Έτσι: $\int_0^2 (\int_0^{4-x^2} (2x)\ dy\ )\ dx \implies \int_0^4 (\int_0^{\sqrt{4-y}} (2x)\ dx\ )\ dy \implies \int_0^4 (\ I\ )\ dy \quad \& \quad I = \int_0^{\sqrt{4-y}} (2x)\ dx$

$\implies I = [x^2]_0^{\sqrt{4-y}} = 4 - y \quad \& \quad \int_0^4 (4-y)\ dy = [4y - \frac{y^2}{2}]_0^4 = \cdots = 16 - 8 = 8$

Τελικά: $\boxed{\int_0^2 \int_0^{4-x^2} 2x\ dy\ dx = 8}$



∗ Χωρίο Ολοκλήρωσης - Σχεδιασμένο στο Geogebra ...

$\underline{\mathbf{3^{η}\ Άσκηση:}}$

$\alpha)$ Δεδομένα: $\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{\left(1+x^2+y^2\right)^2}\, dy\, dx \;=\; \int_{-1}^{1}\Big(\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\Big(\frac{2}{\left(1+x^2+y^2\right)^2}\Big)\, dy\Big)\, dx$

Ξέρουμε τύπους πολικών συντεταγμένων: $x = r\cdot\cos(\theta),\ y = r\cdot\sin(\theta),\ x^2+y^2 = r^2,\ y/x = \tan(\theta)$

Έχουμε: $\begin{cases} \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}}\, dy \;\longrightarrow\; \left\{\begin{array}{c} y=\sqrt{1-x^2} \\ y=-\sqrt{1-x^2}\end{array}\right\} \Longrightarrow\; x^2+y^2=1^2 \;\longrightarrow\; \text{κύκλος ακτίνας } 1 \;\longrightarrow\; \underline{r\in[0,1]} \\ \int_{-1}^{1}\, dx \;\longrightarrow\; \left\{\begin{array}{c} x=1 \\ x=-1\end{array}\right. \;*\; \text{και αφού } r\in[0,1] \;\Longrightarrow\; \underline{\hat{\theta}\in[0,2\pi]} \;*\; \text{και ισχύει } \underline{dxdy \;=\; rdrd\theta} \ ... \end{cases}$

Έτσι: $\displaystyle\int_0^{2\pi}\Big(\int_0^1\Big(\frac{2}{(1+r^2)^2}\cdot r\Big)\, dr\Big)\, d\theta \;=\; 2\cdot\int_0^{2\pi}\Big(\int_0^1\Big(\frac{r}{(r^2+1)^2}\Big)\, dr\Big)\, d\theta$

Θέτω $u = r^2+1 \;\Rightarrow\; u' = \dfrac{du}{dr} \;\Rightarrow\; r\cdot dr = \dfrac{1}{2}\cdot du \;\longrightarrow\; r=0\Rightarrow u=1,\ r=1\Rightarrow u=2.$

$$\Longrightarrow\; \int_0^1 \frac{1}{(r^2+1)^2}\cdot r\, dr = \int_1^2\Big(\frac{1}{u^2}\cdot\frac{1}{2}\Big)\, du = \frac{1}{2}\cdot\Big[-\frac{1}{u}\Big]_1^2 = \frac{1}{2}\Big(\frac{1}{2}\Big) = \frac{1}{4} \;\longrightarrow\; 2\int_0^{2\pi}\Big(\frac{1}{4}\Big)\, d\theta = \frac{1}{2}\cdot(2\pi-0) \;=\; \pi$$

Τελικά: $\boxed{\displaystyle\int_{-1}^{1}\int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{2}{\left(1+x^2+y^2\right)^2}\, dy\, dx \;=\; \pi}$

$\beta)$ Δεδομένα: $z = x^2+y^2 \;\;\&\;\; 0\le x,y\le 2 \;\; ...$

Ξέρουμε: $Avg(f) \;=\; \dfrac{1}{\text{Εμβαδόν χωρίου } R}\cdot\iint_R (f)\, dA \;=\; \dfrac{1}{\text{Εμβαδόν χωρίου } R}\cdot\iint_R f(x,y)\, dx\, dy$

$$\Longrightarrow\; Avg(f) \;=\; \frac{1}{(x_{max}-x_{min})\cdot(y_{max}-y_{min})}\cdot\int_{y_{min}}^{y_{max}}\Big(\int_{x_{min}}^{x_{max}}(z)\, dx\Big)\, dy \;=\; \frac{1}{4}\cdot\int_0^2\Big(\int_0^2(x^2+y^2)\, dx\Big)\, dy$$

$$\Longrightarrow\; \frac{1}{4}\cdot\left(\int_0^2\left(\int_0^2(x^2)\, dx \;+\; \int_0^2(y^2)\, dx\right)\, dy\right) \;=\; \frac{1}{4}\cdot\left(\int_0^2\left(\left[\frac{x^3}{3}\right]_0^2 \;+\; y^2\cdot[x]_0^2\right)\, dy\right) \;=\; \frac{1}{4}\cdot\left(\int_0^2\left(\frac{8}{3}+2y^2\right)\, dy\right)$$

$$\Longrightarrow\; \frac{1}{4}\cdot\left(\int_0^2\Big(\frac{8}{3}\Big)\, dy \;+\int_0^2(2y^2)\, dy\right) \;=\; \frac{1}{4}\cdot\left(\frac{8}{3}\cdot[y]_0^2 \;+2\cdot\left[\frac{y^3}{3}\right]_0^2\right) \;=\; \frac{1}{4}\cdot\left(\;\frac{16}{3}\;+\;\frac{16}{3}\;\right) \;=\; \frac{32}{12} \;=\; \frac{8}{3}$$

Τελικά: $\boxed{Avg(f) \;=\; \dfrac{8}{3}}$

$4^{η}$ Άσκηση:

$\alpha)$ Δεδομένα: στερεό φράσσεται από: $\begin{cases} z = 4 - x^2 \ \ (\text{πάνω}) \\ z = 0 \ \ (\text{κάτω}) \\ x^2 + y^2 = 4 \ \ (\text{πλευρικά}) \end{cases} \longrightarrow d = (4 - x^2) - (0) = 4 - x^2$ $\longrightarrow$ κύκλος με ακτίνα 2 ...

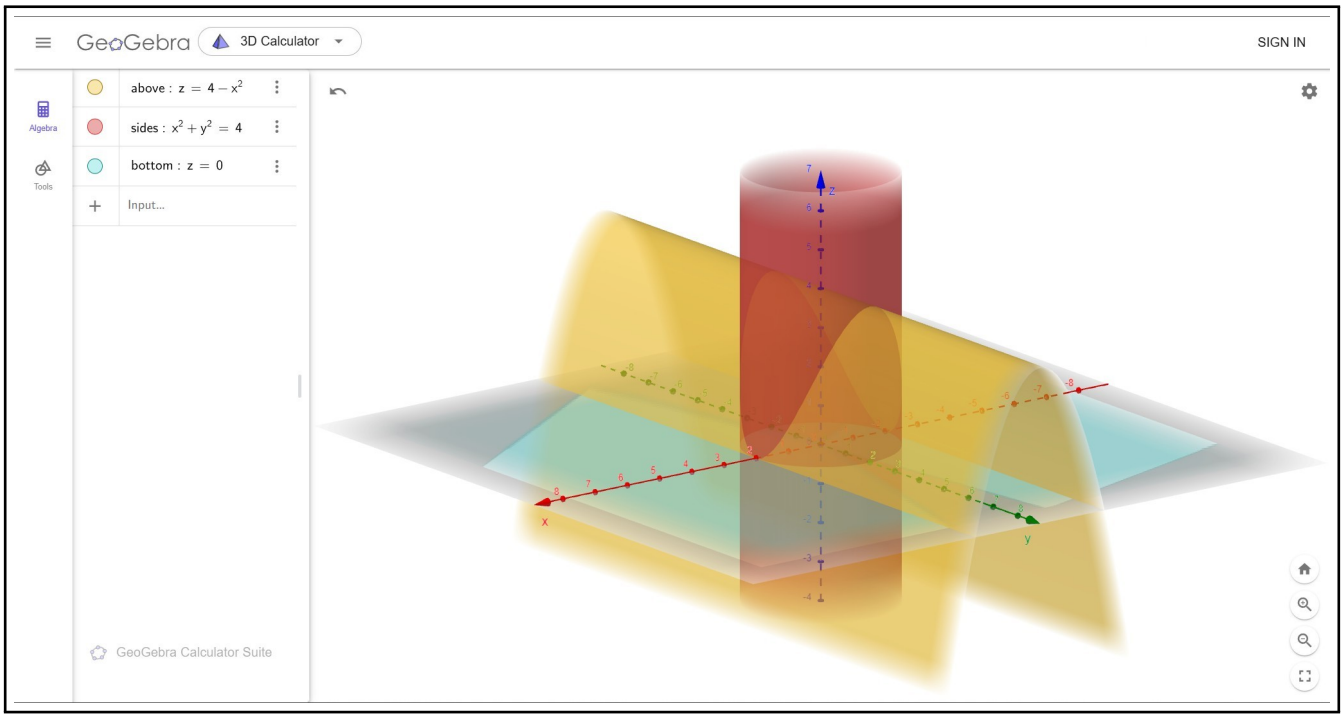
Ξέρουμε: $V = \iint_R dV \implies V = \iint_R d(x,y)\, dxdy$ $\ast$ και $\ast$ Πολικές Συντ. $r \in [0,2]$ & $\theta \in [0, 2\pi]$

$$\ast \quad V = \int_0^{2\pi} \Big(\int_0^2 (4 - r^2\cos^2\theta) \cdot (r)\, dr \Big)\, d\theta = \int_0^{2\pi} \Big(\int_0^2 \left(4r - r^3\cos^2\theta\right)\, dr \Big)\, d\theta = \int_0^{2\pi} \Big(\Big[2r^2 - \frac{r^4}{4}(\cos^2\theta) \Big]_0^2 \Big)\, d\theta$$

$$\implies \int_0^{2\pi} \Big((2\cdot 2^2 - \frac{2^4}{4}(\cos^2\theta)) - (0) \Big)\, d\theta = \int_0^{2\pi} (8 - 4\cos^2\theta)\, d\theta \overset{\cdots}{=} \int_0^{2\pi} \Big(8 - 4 \cdot \Big(\frac{1+\cos(2\theta)}{2} \Big)^{*} \Big)\, d\theta \Rightarrow$$

$$\implies \int_0^{2\pi} (\ 8 - 2 - 2\cos(2\theta))\, d\theta = \int_0^{2\pi} (\ 6 - 2\cos(2\theta))\, d\theta = [6\theta - \sin(2\theta)]_0^{2\pi} = 6(2\pi) - \sin(4\pi) - 0 - (-\sin(0))$$

Τελικά: $\boxed{V = 12\pi}$



$\beta)$ Χωρίο Ολοκλήρωσης - Σχεδιασμένο στο Geogebra ...

5^{η} Άσκηση:

$\alpha)$ Δεδομένα: $\displaystyle\int_0^2\int_0^{4-x^2}\int_0^x \frac{\sin(2z)}{4-z}\,dy\,dz\,dx = \int_0^2\left(\int_0^{4-x^2}\left(\int_0^x\left(\frac{\sin(2z)}{4-z}\right)dy\right)dz\right)dx$

Έχουμε αρχική σειρά: $dy\,dz\,dx \longrightarrow x\in[0,2],\ z\in[0,4-x^2],\ y\in[0,x]$ $\ *\ $ Θα δοκιμάσω: $dy\,dx\,dz\ \ldots$

Είναι: $0\le x\le 2$ και $0\le z\le 4-x^2 \overset{\substack{x=0\\ \Rightarrow\\ x=2}}{} 0\le z\le 4 \Rightarrow z\in[0,4]$

Και: $0\le z\le 4-x^2 \Rightarrow x^2\le 4-z \Rightarrow |x|\le\sqrt{4-z} \overset{0\le x\le 2}{\Rightarrow} x\le\sqrt{4-z} \Rightarrow 0\le x\le\sqrt{4-z} \Rightarrow x\in[0,\sqrt{4-z}]$

Ακόμη: $0\le x\le\sqrt{4-z}$ και $0\le y\le x \Rightarrow 0\le y\le x$

Έτσι: $\displaystyle\int_0^4\int_0^{\sqrt{4-z}}\int_0^x \frac{\sin(2z)}{4-z}\,dy\,dx\,dz = \int_0^4\left(\int_0^{\sqrt{4-z}}\left(\int_0^x\left(\frac{\sin(2z)}{4-z}\right)dy\right)dx\right)dz$

$$\Longrightarrow \int_0^4\left(\frac{\sin(2z)}{4-z}\cdot\left(\int_0^{\sqrt{4-z}}\left(\int_0^x(1)\,dy\right)dx\right)\right)dz$$

$$\Longrightarrow \int_0^4\left(\frac{\sin(2z)}{4-z}\cdot\left(\int_0^{\sqrt{4-z}}(x-0)\,dx\right)\right)dz = \int_0^4\left(\frac{\sin(2z)}{4-z}\cdot\left(\frac{4-z}{2}\right)\right)dz = \int_0^4\left(\frac{\sin(2z)}{\cancel{4-z}}\cdot\frac{\cancel{4-z}}{2}\right)dz$$

$$\Longrightarrow \int_0^4\left(\frac{1}{2}\cdot\sin(2z)\right)dz = \frac{1}{2}\cdot\left[-\frac{\cos(2z)}{2}\right]_0^4 = \frac{1}{2}\cdot\left(-\frac{\cos(8)}{2}+\frac{\cancel{\cos(0)}}{2}^{1}\right) = \frac{1-\cos(8)}{4}$$

Τελικά: $\boxed{\displaystyle\int_0^2\int_0^{4-x^2}\int_0^x \frac{\sin(2z)}{4-z}\,dy\,dz\,dx = \int_0^4\int_0^{\sqrt{4-z}}\int_0^x \frac{\sin(2z)}{4-z}\,dy\,dx\,dz = \frac{\sin^2(4)}{2}}$

6^{η} Άσκηση:

$\alpha)$ Δεδομένα: $y = \sqrt{x},\ \ z = 1 - y,\ 1^{ο}$ οκταμόριο, $dz\ dy\ dx$

$1^{ο}$ οκταμόριο $\longrightarrow\ \ x \geq 0,\ y \geq 0,\ z \geq 0$

$*\ \ z \geq 0\ \ \&\ \ z = 1 - y\ \ \Longrightarrow\ \ 0 \leq z \leq 1 - y\ \ \Longrightarrow\ \ \underline{z \in [0, 1 - y]}$

$*\ \ y \geq 0\ \ \&\ \ y = \sqrt{x}\ \ \&\ \ (\ z \geq 0\ \&\ z \leq 1 - y\ \Rightarrow\ y \leq 1\ )\ \ \Longrightarrow\ \ 0 \leq y \leq \sqrt{x} \leq 1\ \ \Longrightarrow\ \ \underline{y \in [0, \sqrt{x}]}$

$*\ \ x \geq 0\ \ \&\ \ (\ y \leq \sqrt{x}\ \&\ y \leq 1\ \Rightarrow\ x \leq 1^2\ \Rightarrow\ x \leq 1\ )\ \ \Longrightarrow\ \ 0 \leq x \leq 1\ \ \Longrightarrow\ \ \underline{x \in [0, 1]}$

Έτσι: $V\ =\ \iiint\limits_D dV\ =\ \iiint\limits_D F\ dz\ dy\ dx = \int_0^1 \left(\int_0^{\sqrt{x}} \left(\int_0^{1-y} (1)\ dz \right)\ dy \right)\ dx$

$\Longrightarrow\ \int_0^1 \int_0^{\sqrt{x}} (1 - y)\ dy\ dx = \int_0^1 \left[y - \frac{y^2}{2} \right]_0^{\sqrt{x}}\ dx = \int_0^1 \left(\sqrt{x} - \frac{x}{2} \right)\ dx = \left[\frac{2}{3} x^{3/2} - \frac{x^2}{4} \right]_0^1\ =\ \frac{5}{12}$

Τελικά: $\boxed{V_{\alpha}\ =\ \frac{5}{12}}$

$\beta)\ \ \int_0^1 \left(\int_0^{\sqrt{x}} \left(\int_0^{1-y} dz \right)\ dy \right)\ dx\ \ \longrightarrow\ \ \int_?^? \left(\int_?^? \left(\int_?^? dx \right)\ dy \right)\ dz$

Έχουμε: $0 \leq x \leq 1\ \ \&\ \ 0 \leq y \leq \sqrt{x},\ \ \&\ \ 0 \leq z \leq (1 - y)$

$*\ \ 0 \leq x \leq 1\ \ \&\ \ 0 \leq y \leq \sqrt{x}\ \Rightarrow\ 0 \leq y \leq 1\ \ \&\ \ 0 \leq z \leq 1 - y\ \Rightarrow\ 0 \leq z \leq 1\ \ \Longrightarrow\ \ \underline{z \in [0, 1]}$

$*\ \ 0 \leq z \leq 1 - y\ \ \Longrightarrow\ \ 0 \leq y \leq 1 - z\ \ \Longrightarrow\ \ \underline{y \in [0, 1 - z]}$

$*\ \ 0 \leq x \leq 1\ \ \&\ \ 0 \leq y \leq \sqrt{x}\ \Rightarrow\ 0 \leq x \leq 1\ \ \&\ \ 0 \leq y^2 \leq x\ \ \Longrightarrow\ \ y^2 \leq x \leq 1\ \ \Longrightarrow\ \ \underline{x \in [y^2, 1]}$

Τελικά: $\int_0^1 \left(\int_0^{1-z} \left(\int_{y^2}^1 dx \right)\ dy \right)\ dz\ =\ \boxed{\int_0^1 \int_0^{1-z} \int_{y^2}^1 dx\ dy\ dz}$

$\gamma)\ \ \int_0^1 \left(\int_0^{\sqrt{x}} \left(\int_0^{1-y} dz \right)\ dy \right)\ dx\ \ \longrightarrow\ \ \int_?^? \left(\int_?^? \left(\int_?^? dy \right)\ dx \right)\ dz$

Έχουμε: $0 \leq x \leq 1\ \ \&\ \ 0 \leq y \leq \sqrt{x},\ \ \&\ \ 0 \leq z \leq (1 - y)$

$*\ \ 0 \leq x \leq 1\ \ \&\ \ 0 \leq y \leq \sqrt{x}\ \Rightarrow\ 0 \leq y \leq 1\ \ \&\ \ 0 \leq z \leq 1 - y\ \Rightarrow\ 0 \leq z \leq 1\ \ \Longrightarrow\ \ \underline{z \in [0, 1]}$

$*\ \ 0 \leq x \leq 1\ \ \Longrightarrow\ \ x \in [0, 1]$

$*\ \ 0 \leq y \leq \sqrt{x}\ \ \&\ \ 0 \leq z \leq 1 - y\ \Rightarrow\ 0 \leq y \leq \sqrt{x}\ \ \&\ \ 0 \leq y \leq 1 - z\ \ \Longrightarrow\ \ 0 \leq y \leq \min(\sqrt{x},\ 1 - z)$

$\min(\sqrt{x},\ 1 - z)\ \begin{cases} 0 \leq y \leq \sqrt{x}\ \ \Longrightarrow\ \ \sqrt{x} \leq (1 - z)\ \Rightarrow\ 0 \leq x \leq (1 - z)^2\ \ \longrightarrow\ \ \underline{y \in [0,\ \sqrt{x}]\ \ \&\ \ x \in [0,\ (1 - z)^2]} \\ 0 \leq y \leq 1 - z\ \ \Longrightarrow\ \ \sqrt{x} \geq (1 - z)\ \Rightarrow\ (1 - z)^2 \leq x \leq 1\ \ \longrightarrow\ \ \underline{y \in [0,\ 1 - z]\ \ \&\ \ x \in [(1 - z)^2,\ 1]} \end{cases}$

Τελικά: $\boxed{\int_0^1 \int_0^1 \int_0^{\min(\sqrt{x},\ 1-z)} dx\ dy\ dz\ =\ \int_0^1 \int_0^{(1-z)^2} \int_0^{\sqrt{x}} dy\ dx\ dz\ +\ \int_0^1 \int_{(1-z)^2}^1 \int_0^{1-z} dy\ dx\ dz}$

$7^{η}$ Άσκηση:

Δεδομένα: $r = \cos(\theta)$ & $r = 2\cos(\theta)$ & $z = 3 - y$ & $(z = 0)$ $\longrightarrow$ $\iiint_D f(r,\theta,z)\ \mathrm{dz\,dr\,d}\theta$

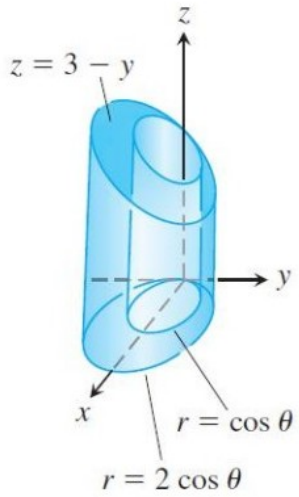
Αρχικά, για τριγύρω έχουμε: $r = \cos(\theta)$ & $r = 2\cos(\theta)$ $\implies$ $\underline{r \in [\cos(\theta),\, 2\cos(\theta)]}$

Επιπλέον, ισχύει: $r \geq 0 \ \Rightarrow\ \cos(\theta) \geq 0 \ \Rightarrow\ -\frac{\pi}{2} + 2k\pi \leq \theta \leq \frac{\pi}{2} + 2k\pi \ \underset{k=0}{\overset{k\in\mathbb{Z}}{\Longrightarrow}}\ -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \implies \underline{\theta \in \left[-\frac{\pi}{2},\, \frac{\pi}{2}\right]}$

Και για πάνω-κάτω είναι: $z = 0$ & $z = 3 - y \ \Rightarrow\ 0 \leq z \leq 3 - y \implies z \in [0,\, 3 - y]$

Αλλά αφού δεν υπάρχει y: $y = r\sin(\theta)$ (κυλινδρικές συντεταγμένες) $\implies$ $\underline{z \in [0,\, 3 - r\sin(\theta)]}$

Τελικά: $\boxed{\int_{-\pi/2}^{\pi/2} \int_{\cos\theta}^{2\cos\theta} \int_{0}^{3-r\sin\theta} f(r,\theta,z)\ \mathrm{dz}\, r\, \mathrm{dr}\, d\theta}$



ΤΕΛΟΣ
(ευχαριστώ)

LaTeX